SMART KNOWLEDGE BASE AGENT

Autonomous Enterprise Hybrid RAG System with Zero-Downtime Indexing

5-Minute Viva & Review Presentation | Department of Computer Science & Engineering

Internal Guide: Ms. M. DEVIPRIYA | HOD: Dr. L. LAKSHMANAN, M.E., Ph.D.

MEMBER 1: SIVA SAI (44110216)

Role: Team Lead & Opening Speaker

Time Allocated: ~1.5 Minutes

- Welcome ma'am & introduce project.
- The Real-World Problem: Manual search is slow; ChatGPT hallucinates & lacks private college data.
- Our Solution: Local AI Agent with cited answers.
- Frontend UI: Streamlit Web UI + Native Desktop App with chat chips & knowledge library table.

MEMBER 2: BHARGAV (44110216)

Role: Architecture & Backend Lead

Time Allocated: ~2 Minutes

- System Architecture Diagram (Fig 4.1):
  Explains all 5 Tiers in simple words.
- Backend API: FastAPI on Port 8000.
- Hybrid Search: Vector Cosine + BM25 Keywords combined with RRF (k=60).
- 0.62 Confidence Gate: Stops fake AI answers if question is out-of-domain.

MEMBER 3: KARTHIKEYA (44110202)

Role: Database & Working Demo Lead

Time Allocated: ~1.5 Minutes

- Dual Database Storage:
  1. ChromaDB: 61 dense 384-dim vectors.
  2. SQLite: metadata.sqlite with SHA-256 hashes.
- How It Works: 5-step operational lifecycle from upload to cited answer.
- Zero Downtime: Background updates without server restarts.
- Live Demonstration Walkthrough & Conclusion.

PROJECT INTRODUCTION & THE PROBLEM STATEMENT

SPEAKER: SIVA SAI (44110216)

SLIDE HIGHLIGHTS (KEY POINTS)

1. What is the Problem?

* In universities and companies, thousands of PDF/DOCX files exist (syllabi, notes, timetables).
* Regular search (Ctrl+F) only looks for exact words and misses conceptual meaning.
* Public AI (ChatGPT) doesn't know our private college documents and often hallucinates (makes up fake facts).
* Cloud AI leaks private data outside the institution.

2. What is our Solution?

* We built the Smart Knowledge Base Agent - a completely local, autonomous RAG system.
* It reads college documents locally on our system.
* Answers student/staff questions in natural English.
* Strictly cites the document name and section so every answer can be verified 100%.

SPOKEN SCRIPT (WHAT SIVA SAI SAYS TO MA'AM)

Speak clearly and confidently (1.5 Minutes):

"Good morning ma'am and respected examiners. I am Siva Sai (44110216), team lead. Together with my teammates Bhargav and Karthikeya, we developed the Smart Knowledge Base Agent.

In our college, we have many PDFs like timetables, 2-mark question banks, and course materials. When students search, normal search fails if they don't type the exact word. If they ask ChatGPT, it doesn't have access to our college files and gives wrong answers.

Our project solves this by creating a private AI assistant that runs right on our system. It reads all our PDFs, understands the meaning of questions, and gives the exact answer with the document citation.

Next, I will explain the Frontend interface that we designed for the users."

THE FRONTEND USER INTERFACE

SPEAKER: SIVA SAI (44110216)

FRONTEND SPECIFICATIONS & SCREENS

Two Modes of Access Built:

1. Web Application: Built using Streamlit, accessible in browser on http://localhost:8501.
2. Native Windows Desktop App: Built using PyWebView (python desktop_app.py). Gives a standalone, borderless Windows software window.

Key UI Components on Screen:

* AI Agent Chat Desk: Interactive search box with query suggestions.
* 1-Click Starter Chips: 'Vuforia & Unity', 'Explain Simply', 'Summarize All' for immediate quick queries.
* Knowledge Library Desk: Shows all 6 indexed files, types (PDF/DOCX), and chunk numbers.
* Controls & Settings: Toggles between local offline synthesis and Gemini AI.

SPOKEN SCRIPT (WHAT SIVA SAI SAYS TO MA'AM)

Spoken Script for Frontend (1 Minute):

"Ma'am, for our frontend, we wanted it to be simple and user-friendly for both teachers and students. We built two interfaces:

First, a Web UI built with Streamlit running on port 8501. Second, a Desktop App using PyWebView that opens as a native Windows window.

On the screen, users have three main tabs:
1. The AI Chat tab, where students can type any question or click quick buttons like 'Vuforia & Unity'.
2. The Knowledge Library tab, which shows all active documents, file sizes, and chunk numbers.
3. The Controls tab, which lets us switch between offline local mode and Gemini cloud mode.

Now, my teammate Bhargav will explain our System Architecture Diagram and the Backend engine."

SYSTEM ARCHITECTURE DIAGRAM (FIGURE 4.1)

SPEAKER: BHARGAV (44110216)

SYSTEM ARCHITECTURE DIAGRAM (FIVE-TIER DESIGN)

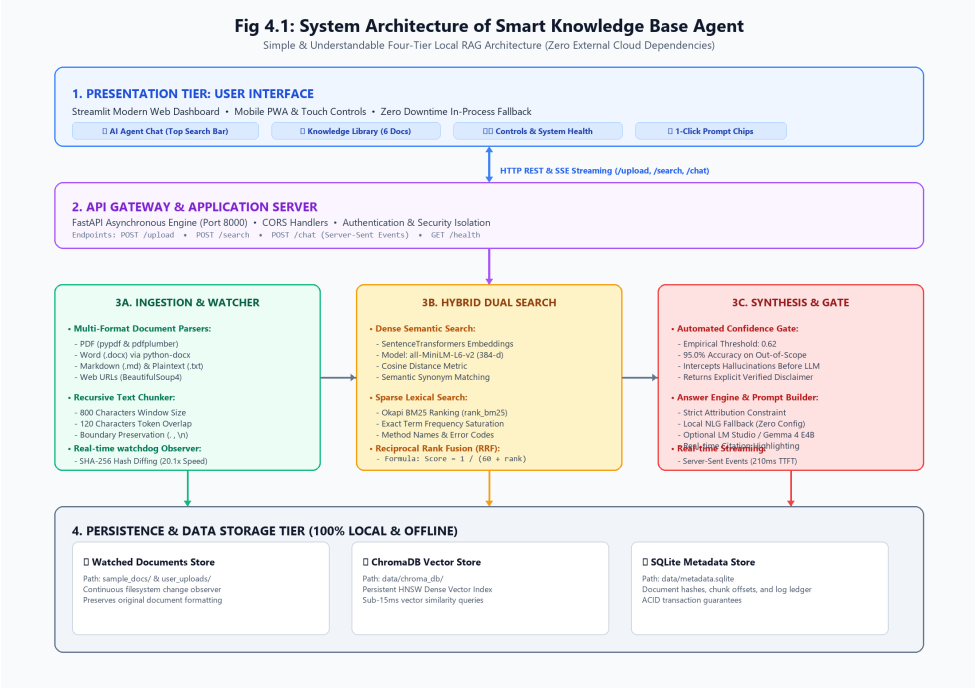


Figure 4.1: Five-Tier System Architecture of Smart Knowledge Base Agent

SPOKEN SCRIPT (WHAT BHARGAV SAYS TO MA'AM)

Explaining the Diagram in Simple English (1.5 Minutes):

"Good morning ma'am. I am Bhargav (44110216). I will explain our System Architecture Diagram.

As shown in Figure 4.1, our project is organized into 5 simple, connected tiers:

* Tier 1 - Presentation Layer: This is the Streamlit screen where the user inputs questions.
* Tier 2 - API Routing Layer: Powered by FastAPI on port 8000. It receives the question and sends it to our RAG engine.
* Tier 3 - Hybrid RAG Engine: This is the brain. It runs Vector Search and BM25 Keyword Search together, merges the results using RRF (k=60), and checks the 0.62 Confidence Gate.
* Tier 4 - Dual Database: ChromaDB stores 61 vector embeddings, while SQLite stores file hashes and timestamps.
* Tier 5 - Background Ingestion: A continuous scanner that watches our uploads folder and auto-indexes new files without stopping the server.

Next, I will explain how our Backend Search works."

THE BACKEND ENGINE & HYBRID SEARCH

SPEAKER: BHARGAV (44110216)

BACKEND & SEARCH SPECIFICATIONS

1. FastAPI Server (main.py):

* Runs on Uvicorn on Port 8000.
* Provides 7 REST API routes (/chat, /search, /upload, /documents, /health).
* Includes live Swagger UI testing at http://localhost:8000/docs.

2. Hybrid Search (Vector + BM25):

* Vector Search: Uses all-MiniLM-L6-v2 (384 dimensions) to understand meaning.
* BM25 Search: Finds exact keywords, roll numbers, course codes.
* Reciprocal Rank Fusion (k=60): Combines both scores fairly without bias.

3. Confidence Gate (Threshold = 0.62):

* Prevents AI hallucinations. If top similarity is < 0.62, the system safely says 'Not found in knowledge base'.

SPOKEN SCRIPT (WHAT BHARGAV SAYS TO MA'AM)

Backend Spoken Script (1 Minute):

"Ma'am, for our backend, we used FastAPI with Python on port 8000.

The most important technical feature here is our Hybrid Search. If you do only Vector search, it might miss exact words like 'Vuforia' or roll numbers. If you do only Keyword search, it misses questions written in different words.

So we run both Vector search and BM25 search in parallel, and combine them using Reciprocal Rank Fusion with constant k=60.

We also added a 0.62 Confidence Gate. If someone asks a question outside our documents, like 'How to drive a car', the confidence is below 0.62. Our system politely says 'Information not found' instead of inventing fake answers.

Now, my teammate Karthikeya will explain our Database layer and demonstrate how the project works."

THE DUAL DATABASE STORAGE (CHROMADB + SQLITE)

SPEAKER: KARTHIKEYA (44110202)

DATABASE ARCHITECTURE & FILES

Why Dual Database (Two Databases)?

A relational database cannot do fast vector search, and a vector database is not built for relational metadata. So we separate concerns:

1. ChromaDB (Vector Store): data/chroma_db

* Stores 61 dense vector embeddings (384 numbers per chunk).
* Uses HNSW (Hierarchical Navigable Small World) index for sub-millisecond nearest neighbor search.

2. SQLite Database: data/metadata.sqlite

* Tracks 6 documents in table 'files'.
* Stores SHA-256 cryptographic hashes and upload timestamps.
* Skips re-indexing unchanged documents in < 0.04s.

SPOKEN SCRIPT (WHAT KARTHIKEYA SAYS TO MA'AM)

Database Spoken Script (1 Minute):

HOW THE PROJECT WORKS & LIVE DEMO

SPEAKER: KARTHIKEYA (44110202)

5-STEP WORKING LIFECYCLE & LIVE DEMO

5-Step Operational Lifecycle:

1. Document Ingestion: PDF/Word files added to data/uploads/.
2. Hash Check: SQLite checks SHA-256 hash to prevent duplicates.
3. Text Chunking: Text split into 500-token chunks with 50-token overlap.
4. Embedding: Chunks converted to 384-dim vectors and saved to ChromaDB.
5. Gated Retrieval: User asks query -> Hybrid search retrieves top chunks -> Gated check -> Grounded answer generated.

What to Show Ma'am During Demo:

* 1. Run Frontend: Double-click run_app.bat -> Click 'Vuforia & Unity' -> Show cited answer.
* 2. Run Backend Docs: Open http://localhost:8000/docs -> Test GET /health.
* 3. Run Database Audit: In PowerShell run .\venv\Scripts\python.exe show_db.py.

SPOKEN SCRIPT & CLOSING (KARTHIKEYA)

Closing Script (1 Minute):

"Ma'am, here is how the whole project works from start to finish:

When an administrator drops a new syllabus PDF into our uploads folder, our Directory Scanner picks it up immediately. It splits the document into 500-token chunks, creates 384-dimensional vectors, and saves them into ChromaDB.

The best part is Zero Downtime: new files are indexed in the background without restarting the server.

Now, let us show you the live running project:

1. We run run_app.bat to launch our frontend.
2. We click the 'Vuforia & Unity' button on screen to show the exact answer from our college document.
3. We show our Swagger API at port 8000.
4. And we run python show_db.py in terminal to prove our 61 vectors and 6 files in ChromaDB and SQLite.

Thank you ma'am! We are now happy to answer your questions."

VIVA CHEAT-SHEET - QUICK ANSWERS FOR MA'AM

ALL 3 MEMBERS PREPARATION

Q1: Why did you use two databases instead of one?

"Ma'am, SQLite is an ACID relational database perfect for file metadata and fast SHA-256 hash checking. ChromaDB is a specialized vector database with HNSW graph indexing for 384-dimensional semantic search. Using both gives optimal speed and clean separation."

Q2: What is the 0.62 Confidence Gate and why is it needed?

"Ma'am, it stops AI hallucinations. If someone asks a question not in our documents, the similarity score is below 0.62. The system safely refuses to answer instead of making up false information."

Q3: What is RRF (Reciprocal Rank Fusion, k=60)?

"Ma'am, it is a mathematical formula that merges Vector Search results and BM25 Keyword Search results fairly, without needing manual weight tuning."

Q4: What is Zero Downtime Indexing?

"Ma'am, when a new PDF is added, our Directory Scanner indexes it in the background. The user can keep chatting and searching without turning off or restarting the server."

Q5: What embedding model are you using?

"We use sentence-transformers/all-MiniLM-L6-v2. It generates 384-dimensional dense vectors and runs fast locally on CPU in under 15 milliseconds per chunk."